

COURSE DESCRIPTION: EE-3009 Computer Architecture (CA)

COURSE DESCRIPTION FORM

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Karachi

PROGRAM TO BE EVALUATED

BS-School of Computing– Fall 2026

Course Description

Course Code	EE-3009		
Course Title	Computer Architecture		
Credit Hours	3		
Prerequisites by Course(s) and Topics	DLD,COAL		
Grading Policy	Absolute grading		
Policy about missed assessment items in the course	Retake of missed assessment items (other than midterm/ final exam) will not be held. For a missed midterm/ final exam, an exam re-take/ pre-take application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee will decide the exam re-take/ pre-take cases.		
Course Plagiarism Policy	Plagiarism in project or midterm/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.		
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Assessment Item	Number	Weight (%)
	Assignment	2	8%
	Quiz	4	12%
	Midterm Exam	2	30%
	Final Exam	1	50%
Course Instructors	Aashir Mahboob, Kashan Hussain , Dr.Noman Durrani,Abeeha Sattar ,Atiya Jokhio & Abdullah Saud Fatmi		
Lab Instructors (if any)			
Course Coordinator	Aashir Mahboob		
URL (if any)			

Current Catalog Description	- Get a working knowledge of architecture of subsystems and the general principles that affect their performance - Get knowledge about advanced architectural features that boost the performance of computers with pipelining and superscalar architecture - Analyze the performance of systems and quantify the performance measurements - Study the major trends towards parallelism with respect to a single processor and multicore architecture
Textbook	Computer Architecture: A Quantitative Approach <i>John L. Hennessy & David A. Patterson 6th Edition</i>
Reference Books	Computer Organization & Architecture <i>William Stallings Pearson, 9th Edition</i>

S#	CLO	Domain	Taxonomy level	PLO
1	Describe the performance evaluation criteria of computers and recognize performance of different computing systems.	Cognitive	2	1
2	Describe advanced concepts in computer architecture like multi-cycle pipeline, dynamic scheduling, dynamic branch prediction, loop unrolling, super-scalar, multi-issue and multi-core processors and identify their impact on a given machine.	Cognitive	2	1
3	Analyze existing bottlenecks in computer architecture designs and categorize potential solutions, virtual memory contribution toward performance improvement of a computing machine.	Cognitive	4	2

Program Learning Outcome (PLO)	Description
	PLO-01: Computing Knowledge: An ability to apply knowledge of mathematics, science, computing fundamentals and an engineering specialization to the solution of complex computing problems.
	PLO-02: Problem Analysis: Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.

	Topics to be covered				
	List of Topics	Week	Chapter/ Appendix	Contact Hours	CLO(s)
Topics covered in the course with number of lectures on each topic (Assume 16 weeks of instruction and 1 hour lecture duration)	Introduction (1 Lecture) -----	1	1	3	1
	Classes of Computers (1 Lecture) -----				
	Performance Metrics of a system (1 Lecture)				
	Measuring and Reporting Performance (1 Lecture) -----	2	1	3	1
	Amdahl Law (1 Lecture) -----				
	CPU Performance Equation (1 Lecture)				
	----- Chapter related fallacies and pitfalls (1 Lecture) -----	3	1 & A	3	1
	Principals of ISA, internal Storage, Memory addressing (2 Lecture) -----				
	Control Flow Instructions, Role of Compilers, IS encoding (1.5 Lecture) -----	4	A	3	3
	Register Allocation (1.5 Lecture)				

	MIPS Architecture (2 Lecture) ----- Instruction Usage , Chapter related fallacies and pitfalls (1 Lecture)	5	A	3	2
	WEEK 6	MID -1 Exam			
	Pipelining Basic concepts, limitations, Major Hazards of pipelining & Internal Forwarding (3 Lectures)	7	C	3	3
	Exception in Pipeline, Precise Exceptions & Floating-point Pipeline (3 Lectures)	8	C	3	3
	MIPS R-4000 Pipeline, Superscalar and VLIW architecture. (3 Lectures)	9	C	3	2

		Instruction Level Parallelism, data and name dependance, Loop unrolling (3 Lectures)	10	3	3	3
	Week 11	MID -2 Exam				
		Dynamic Scheduling -Scoreboarding technique and Tomasulo's Approach (3 Lectures)	12	3&C	3	1
		Basics of Caches, Caches miss , hits & Organization (3 Lectures)	13	B	3	3
		Four memory hierarchy questions, Six basic cache optimizations (3 Lectures)	14	B	3	3
		Advance Cache Optimizations, Virtual memory , DRAM optimizations (3 Lectures)	15	B	3	3



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	Simultaneous Multithreading (SMT), Multiprocessors Thread-Level parallelism (3 Lectures)	16	5	3	3
	<i>Total</i>		16	48	
Laboratory Projects/Experiments Done in the Course	None.				
Programming Assignments Done in the Course	None.				
Class Time Spent (in percentage)	Theory (%)	Problem Analysis (%)	Solution Design (%)	Social and Ethical Issues (%)	
	50	25	20	5	

Instructor Name: Aashir Mahboob

Instructor Signature: Aashir Mahboob

Date: 5th-August-2026